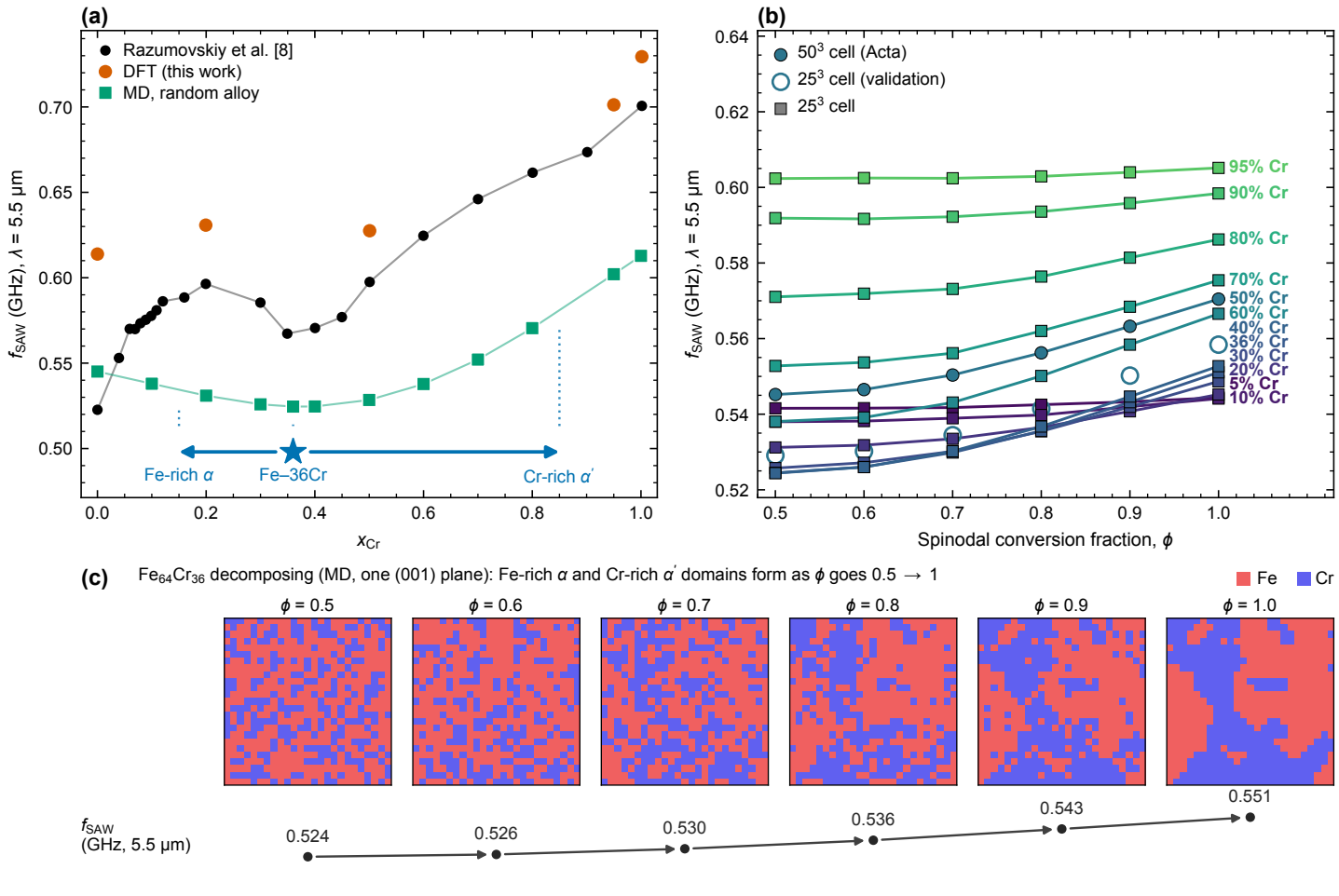

(a)
Razumovskiy et al. [8]
DFT (this work)
MD, random alloy
$f_{SAW}$ (GHz), $\lambda$ = 5.5 µm
$x_{Cr}$
Fe-rich $\alpha$
Fe–36Cr
Cr-rich $\alpha'$
(b)
$50^3$ cell (Acta)
$25^3$ cell (validation)
$25^3$ cell
95% Cr
90% Cr
80% Cr
70% Cr
50% Cr
60% Cr
40% Cr
36% Cr
30% Cr
20% Cr
5% Cr
10% Cr
Spinodal conversion fraction, $\phi$
(c)
$Fe_{64}Cr_{36}$ decomposing (MD, one (001) plane): Fe-rich $\alpha$ and Cr-rich $\alpha'$ domains form as $\phi$ goes 0.5 → 1
Fe
Cr
$\phi$ = 0.5
$\phi$ = 0.6
$\phi$ = 0.7
$\phi$ = 0.8
$\phi$ = 0.9
$\phi$ = 1.0
$f_{SAW}$ (GHz, 5.5 µm)
0.524
0.526
0.530
0.536
0.543
0.551